

LAQUAL: An Automated Framework for LLM App Quality Evaluation

Yan Wang^a, Xinyi Hou^b, Junjun Si^{a,*}, Yanjie Zhao^b, Weiguo Lin^a, Haoyu Wang^b

^a*School of Computer and Cyber Security, Communication University of
China, Beijing, China*

^b*School of Cyber Science and Engineering, Huazhong University of Science and
Technology, Wuhan, China*

Abstract

Representing a new paradigm in software distribution, LLM app stores are rapidly emerging, offering users diverse choices for content generation, coding assistance, education, and more. However, current ranking and recommendation mechanisms in LLM app stores predominantly rely on static metrics, such as user interactions and favorites, making it challenging for users to efficiently identify high-quality apps. At the same time, current academic research focuses on specific vertical fields and lacks a general, automated evaluation framework applicable to the diverse LLM app ecosystem. To address the above challenges, we present LAQUAL, an automated framework for LLM app quality evaluation. LAQUAL integrates three key stages: (1) LLM app labeling and hierarchical classification for precise scenario mapping; (2) static indicator evaluation using time-weighted user engagement and functional capability indicators to filter low-quality apps; and (3) dynamic scenario-adaptive evaluation, where an LLM generates scenario-specific evaluation metrics, scoring criteria, and tasks for comprehensive quality evaluation. Experiments on a mainstream LLM app store demonstrate the effectiveness of LAQUAL. Its automated scores show high consistency with human judgments (Spearman’s $\rho = 0.62$, $p = 0.006$ in legal consulting; $\rho = 0.60$, $p = 0.009$ in travel planning). Through effective screening, LAQUAL can reduce the candidate LLM app pool by 66.7% to 81.3%. User studies further validate its significant outperformance over baseline systems, particularly in comparison efficiency (mean 5.45 vs. 3.30) and value of explanatory information (4.75 vs. 2.25). These results demonstrate that LAQUAL provides a

*The corresponding author is Junjun Si (sijunjun@cuc.edu.cn).

scalable, objective, and user-centric solution for high-quality discovery and recommendation of LLM apps in real-world scenarios.

Keywords:

LLM app stores, Quality evaluation, Automated evaluation framework

1. Introduction

Large language models (LLMs) have made breakthrough progress in recent years, which has spurred the emergence of a wide range of LLM-based intelligent software, such as LLM apps, LLM agents, self-hosted LLM services, and edge-LLM applications. Among these software forms, this study focuses on **LLM apps** [1], which specifically refer to intelligent applications built upon LLM capabilities and available on various **LLM app stores** such as OpenAI’s GPT Store [2], Quora’s Poe [3], Baidu’s Smart Cloud Qianfan AppBuilder (AppBuilder) [4], and Tencent’s Yuanqi Platform [5]. These LLM apps typically feature independent entry points, functional descriptions, and user interfaces, and are widely used in scenarios such as text generation, content creation, education, coding assistance, and emotional support [1, 3, 6, 7]. Compared to traditional apps, LLM apps not only understand and generate natural language but also support complex capabilities such as external knowledge base integration, plugin invocation, and multimodal input, significantly expanding the boundaries of artificial intelligence (AI) applications [8, 9, 10, 11]. Driven by rapid technological advances and lowered development barriers, the LLM app ecosystem has experienced explosive growth, with mainstream LLM app stores hosting a vast number of LLM apps. In just a few months since its launch, the GPT Store has already included over 3 million GPT apps [12], FlowGPT has over 4 million monthly active users [13], and platforms like Poe, and HuggingChat have each exceeded 10 million monthly visits [14, 15]. Furthermore, third-party platforms such as GPTStore.AI [16] and GPTs Hunter [17] have also emerged, with the app inventory on a single platform reaching hundreds of thousands, demonstrating explosive growth in overall market size and activity.

This rapid expansion brings new challenges for users in discovering and selecting high-quality LLM apps. On the one hand, the surge in the number of apps has led to information overload, making it difficult for users to find high-quality apps that truly meet their needs efficiently. On the other hand, current mainstream LLM app stores generally adhere to traditional

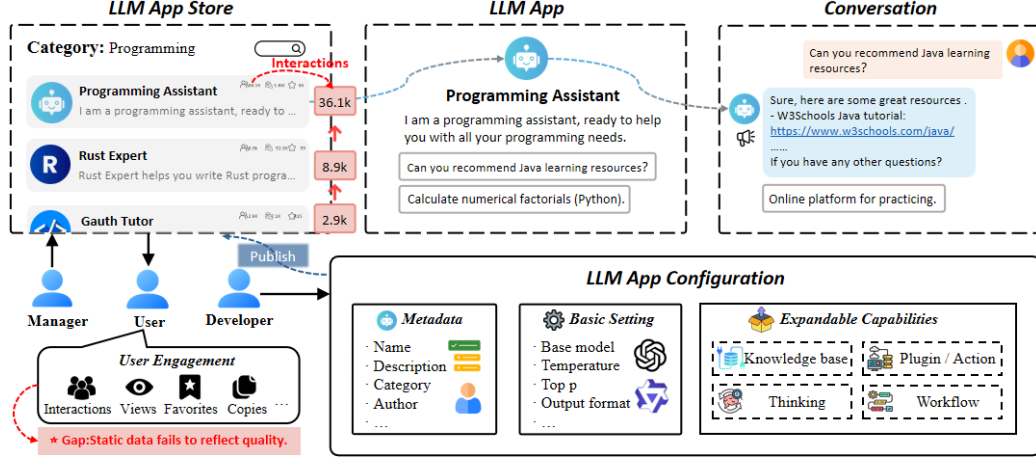


Figure 1: Overview of the LLM App Ecosystem.

64 recommendation logic: users input searching keywords, and the platform
 65 then returns a list of apps ranked by static metrics such as usage, number of
 66 favorites, and release date [18, 19]. With this scheme, users must manually
 67 filter and repeatedly try out various apps, incurring significant cognitive and
 68 time costs. In addition, recent user discussions have revealed that app rank-
 69 ings are frequently manipulated through methods such as keyword stuffing,
 70 fake reviews, and misleading updates, resulting in low-value apps appearing
 71 at the top of search results [20, 21, 22]. In summary, as illustrated in Figure 1,
 72 the current mechanism, which relies on static metrics, makes it difficult for
 73 LLM App users to accurately identify truly high-quality apps, thus severely
 74 damaging the user experience and the health of the ecosystem.

75 Although LLM evaluation methods evolve, they typically focus on specific
 76 domains, such as law, mathematics, or code generation, and rely on corre-
 77 sponding benchmarks with predefined metrics [23, 24, 25, 26, 27]. While these
 78 benchmarks are effective for evaluating LLMs in well-defined scenarios, it’s
 79 difficult to adapt them to the LLM app store context. The diversity of app
 80 categories can reach dozens or even more, making a comprehensive evalua-
 81 tion of various types of apps impractical using existing methods [28, 29]. Fur-
 82 thermore, unlike general-purpose or domain-specific LLMs, individual LLM
 83 apps often target highly specialized or niche functionalities, which require
 84 evaluation at a much finer granularity to accurately reflect their value and
 85 effectiveness. Most importantly, the ultimate goal of LLM app stores is to
 86 serve users; therefore, factors such as user engagement, experience, and sat-

isfaction must be fully considered in the evaluation process. However, the above benchmarks are mostly model-centric and static, lacking mechanisms to capture real user interactions or feedback. Therefore, a new evaluation framework that can adapt to the multidimensional, fine-grained, and user-centric nature of the LLM app ecosystem is urgently needed.

In response to these unmet needs, we propose LAQUAL, an automated framework for LLM app quality evaluation, designed to be broadly applicable across diverse LLM app scenarios. LAQUAL first conducts scenario-aware labeling and classification to accurately map LLM apps to their intended use cases. Next, it performs multi-dimensional static filtering based on time-weighted user engagement and functional capability indicators to efficiently exclude low-quality candidates. Finally, LAQUAL employs LLM-driven dynamic assessment, automatically generating scenario-adaptive evaluation metrics and tasks to enable fine-grained automatic LLM app quality measurement.

We systematically evaluate the effectiveness of our framework from three key perspectives: evaluation accuracy and adaptivity, recommendation effectiveness, and user impact. First, we assess whether LAQUAL can provide accurate and scenario-adaptive quality evaluation by comparing its automated scores with human judgments of 28 human evaluators (including domain experts) across different LLM app scenarios. Second, we benchmark LAQUAL against mainstream app store recommendation mechanisms, demonstrating that our framework substantially reduces the candidate app pool while surfacing higher-quality apps. Third, we examine the impact of LAQUAL on users’ decision quality, cognitive effort, and trust in real-world app discovery tasks through a controlled user study with 12 participants ($N = 12$) using a within-subjects design. Results show that LAQUAL improves decision support and comparison efficiency for users, and increases the perceived value of explanatory information compared to baseline systems. Collectively, these evaluations validate the practical effectiveness, rationality, and user-centered benefits of our solution in real-world LLM app scenarios.

In summary, this work makes the following key contributions:

- We design and implement LAQUAL, an automated framework featuring a multi-stage pipeline for scalable and systematic LLM app quality evaluation, achieving both efficient and accurate evaluation results.
- We propose an efficient static indicator evaluation module that utilizes time-weighted user engagement and functional capability indicators,

reducing low-quality app candidates by 66.7%–81.3% and significantly improving evaluation efficiency.

- We develop a fully automated dynamic evaluation pipeline, where LLMs are employed to generate scenario-specific evaluation metrics and tasks, enabling fine-grained, context-adaptive, and objective quality evaluation across diverse app categories, achieving strong consistency with human judgments (Spearman’s $\rho = 0.62$, $p = 0.006$ in legal consulting, and $\rho = 0.60$, $p = 0.009$ in travel planning).
- We conduct comprehensive real-world validation, showing that LAQUAL achieves high agreement with expert ratings, and substantially boosts user decision confidence (mean 5.05 vs. 4.40), comparison efficiency (5.45 vs. 3.30), and perceived value of explanatory information (4.75 vs. 2.25) compared to baseline systems.

The remainder of this paper is organized as follows. § 2 reviews relevant background on LLM apps and LLM-based evaluation. § 3 introduces the architecture and key components of LAQUAL. In § 4 validates the framework’s effectiveness through experiments and user studies. § 5 discusses the main findings and their broader significance. § 6 outlines current limitations and future directions. § 7 summarizes related work, and § 8 concludes the paper.

2. Background

2.1. LLM App

LLM apps are specialized applications built on LLMs that provide intelligent, task-oriented services through natural language interaction [1]. Unlike traditional software, LLM apps leverage the reasoning, comprehension, and generation abilities of foundation models to support diverse scenarios, such as content creation, code assistance, consultation, translation, and data analysis. LLM apps include metadata like app name, description, instructions, categories, developer details, and sample conversations, helping users understand each app’s core features and use cases.

LLM app stores are digital platforms designed for the aggregation, distribution, and management of various LLM apps. Notable examples include GPT Store [2], Coze [6], and Poe [3]. These platforms provide developers with efficient tools for publishing, configuring, and hosting LLM apps, as

157 well as supporting streamlined creation, deployment, and maintenance work-
158 flows [1]. Recalling the structural components in Figure 1, developers can
159 incorporate functionalities such as knowledge bases, plugins, and thinking,
160 while enriching app metadata to improve clarity and usability. For users,
161 LLM app stores offer intuitive interfaces to search, browse, and discover
162 apps by categories or tags. Users can interact with selected apps through
163 multi-turn natural language conversations, experiencing a variety of intelli-
164 gent services. The platforms also collect and utilize user behavioral data and
165 feedback, including ratings, usage statistics, and number of conversations, as
166 important factors in app recommendation and ranking.

167 In recent years, the low technical barrier for LLM app development has led
168 to an explosive growth in the number of available apps on these platforms.
169 This rapid proliferation creates significant challenges for users, who often
170 struggle to efficiently identify and select high-quality apps that best meet
171 their needs. Addressing this challenge and enabling effective discovery and
172 evaluation of LLM apps is a critical motivation for our work.

173 2.2. LLM-based Evaluation Paradigms

174 Traditional evaluation methods often lack flexibility and semantic un-
175 derstanding when assessing complex language outputs. To address this,
176 researchers have begun using LLMs as evaluators, a paradigm known as
177 **LLM-as-a-Judge**. This approach leverages the LLM’s reasoning abilities to
178 overcome the rigidity, limited scalability, and semantic shortcomings of tradi-
179 tional methods. A line of pioneering studies has demonstrated the potential
180 of this paradigm. For example, Chiang et al. [30] showed that LLM-based
181 evaluations are often highly consistent with expert human judgment, espe-
182 cially in open-ended tasks like story generation. In the education domain,
183 Chang et al. [31] and Grévisse et al. [32] leveraged GPT-family and Gem-
184 ini models, respectively, for automatic grading of short-answer questions,
185 achieving promising results in terms of accuracy and efficiency compared to
186 human annotators. These advances indicate that LLM-as-a-Judge could sub-
187 stantially reduce labor costs and improve handling of edge cases in various
188 vertical applications.

189 Nevertheless, a growing body of research also reveals significant challenges
190 in generalizing this approach. As summarized by Gu et al. [33] and others,
191 the reliability and stability of LLM-based evaluation are often sensitive to
192 prompt design and model sampling parameters. Furthermore, many current
193 studies rely on limited or single-source test sets and lack systematic scoring

standards, which makes it difficult to extend their findings to diverse, real-world LLM application scenarios involving complex interactions.

Motivated by these gaps, this work presents the LAQUAL framework, which systematically optimizes the LLM-as-a-Judge paradigm and, for the first time, applies it to the comprehensive evaluation of complex LLM apps. To overcome the core limitations of previous approaches, particularly the lack of explicit evaluation criteria and insufficiently diverse test sets, LAQUAL introduces a dynamic generation mechanism that automatically produces task-specific scoring metrics and evaluation questions before assessment.

3. Proposed framework

This section introduces the workflow of LAQUAL. As shown in Figure 2, LAQUAL consists of three key stages: *LLM app labeling and classification* (§ 3.1), *static indicator evaluation* (§ 3.2), and *dynamic scenario-adaptive evaluation* (§ 3.3).

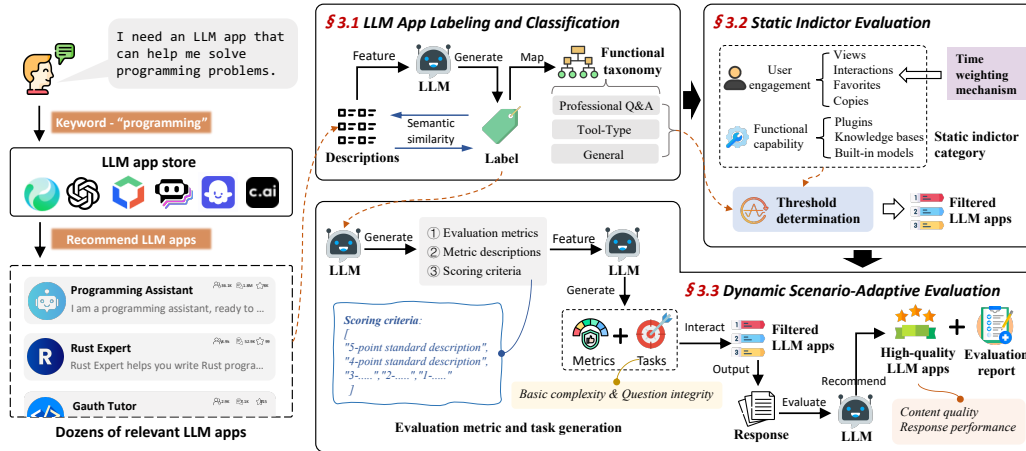


Figure 2: The Workflow of LAQUAL.

3.1. LLM App Labeling and Classification

This subsection is the foundation of LAQUAL, transforming unstructured metadata into standardized semantic representations for categorization. Based on raw app descriptions, the system automatically generates labels via an LLM-based annotator. These labels are then mapped to a hierarchical taxonomy, ensuring a basis for the subsequent evaluation.

214 3.1.1. LLM-based Labeling

215 Before the formal evaluation begins, an *LLM-based annotator* automati-
 216 cally generates labels for all LLM apps to be evaluated. In LLM app stores,
 217 each app is associated with a description. The LLM-based annotator per-
 218 forms semantic analysis on this textual information to extract key domain
 219 entities, functional elements, and typical usage scenarios. Based on this
 220 analysis, the annotator generates a candidate label that summarizes the core
 221 functionality of the app. To ensure that labels are concise and accurate,
 222 LAQUAL limits their length. For example, if it is Chinese, it can be lim-
 223 ited to a phrase containing 6-9 Chinese characters. Figure 3 illustrates the
 224 prompt of the LLM-based annotator.

225 We then use the *text2vec* model to further verify the accuracy of the
 226 generated labels. Specifically, this model calculates the semantic similarity
 227 between each candidate label and the original description of the LLM app.
 228 A similarity threshold of 0.7 is set; if the calculated similarity falls below
 229 this threshold, a feedback mechanism is immediately triggered, prompting
 230 the LLM to iteratively refine the label until the similarity requirement is met
 231 or exceeded. It is important to note that the threshold of 0.7 is empirically
 232 determined, based on extensive experimental validation and a comprehen-
 233 sive assessment of both label quality and iteration efficiency under different
 234 thresholds.

```

As an LLM-based annotator, please analyze the following app descriptions, extract the core entities, and generate a label that
best highlights the core value and characteristics of these applications.
Application Description:
{json.dumps(descriptions, ensure_ascii=False, indent=2)}

Analysis Steps:
1. Extract Core Entities
The label needs to extract the core areas and core functions in the description
2. Label requirements
Must be concise (6-9 words), professional and specific
3. Banned Word List
To ensure the specificity of the label, the following high-frequency but vague common words must not appear in the final
generated label:[.....]
4. Examples:<examples>

```

Figure 3: Prompt Template for LLM-based Annotator.

235 To avoid ambiguity, we explicitly distinguish between *labels* and *tags* in
 236 LAQUAL. A *label* is a concise semantic summary automatically generated
 237 from the app description, whereas a *tag* is a predefined taxonomy item used

in the subsequent hierarchical classification stage.

3.1.2. Hierarchical Classification

After generating labels for each LLM app, the next step is to determine its specific category. As illustrated in Table 1, we designed a three-level hierarchical taxonomy, consisting of *category*, *subcategory*, and *tag*, to categorize LLM apps based on their core functional characteristics.

Table 1: Functional Taxonomy of LLM Apps.

Category	Subcategory	Tags
Professional Q&A	Medical Health	Medical, Health, Disease, Diagnosis, Treatment, Psychology
	Legal Consultation	Law, Regulation, Contract, Litigation
	Financial Management	Finance, Financial Management, Investing, Stocks
	Education	Education, Training, Courses, Mathematics, Chinese, Biology, Physics, Chemistry, History, Geography, Politics
	Language Learning	Translation, Vocabulary, Translation, Grammar, Language, Chinese, English, Japanese, German, Korean, French, Russian, Italian
	Professional Consulting	Consulting, Consultant, Expert
Tool-type	Development Tools	Code, Programming, Development, Debugging, Algorithms, Front-end, Back-end, Python, Java, Javascript, Matlab
	Analysis Tools	Data Analysis, Statistical Analysis, Data Forecasting
	Planning Tools	Planning, Planner, Planning Design
	Design Tools	Interior Design, Branding
General	–	–

The taxonomy was constructed using a mixed approach that combined abstraction informed by prior work with corpus-based induction. Specifically, we used the classification results reported in Su et al. [34] as an initial reference and manually abstracted candidate functional categories. Semantically similar or overlapping categories were then iteratively compared and merged through multiple rounds of refinement, resulting in 10 subcategories and 3 top-level categories: *professional Q&A*, *tool-type*, and *general*. For the fine-grained layer, the 62 tags were derived from high-frequency functional, usage-scenario, and domain-specific terms extracted from the names, descriptions, and related metadata of more than 2,000 apps collected from the Coze and Baidu Qianfan platforms. These candidate tags were then further refined through duplicate removal, synonym merging, and ambiguity filtering.

Rather than directly mapping the app descriptions to the predefined classification tags, LAQUAL instead maps these tags using the generated labels

in the previous step. **These labels serve as a semantic bridge, enabling more accurate and robust mapping to our fine-grained taxonomy by reducing the noise and ambiguity of the original descriptions.** LAQUAL employs a hierarchical matching strategy, from fine-grained to coarse-grained, prioritizing tag over subcategory. In particular, LAQUAL first attempts to match label with the most specific tag keywords, such as “litigation” and “contract” under the “Legal Consultation” subcategory, to ensure precise classification. If no match is achieved at the tag level, the process falls back to broader subcategory keywords, such as “Education” or “financial management”, for subsequent matching. This hierarchical retrieval approach not only ensures classification accuracy but also significantly improves the efficiency and robustness of the identification process.

3.2. Static Indicator Evaluation

As the preliminary screening layer of LAQUAL, this subsection evaluates LLM apps through multi-dimensional static platform data. This process first identifies key interaction and capability indicators, then applies a time-decay weighting mechanism to ensure fairness across apps with different release dates.

3.2.1. Static Indicator Selection

The first step in static indicator evaluation is to determine the scope of indicators to consider. We selected two-dimensional evaluation indicators based on objective and readily available platform data. Our goal is to provide a comprehensive evaluation that reflects both user interaction patterns and the inherent technical capabilities of each LLM app.

User engagement indicators, which measure the app’s popularity and recognition among the user community; and **functional capability indicators**, which assess the app’s configuration richness and versatility. This dual-perspective approach enables us to distinguish between apps that attract broad user attention and those that offer advanced technical features. Table 2 lists the detailed indicators used in our evaluation framework. For example, *copies*, or configuration replication frequency, reflects an app’s influence among developers, while *plugins* indicates its extensibility and the ability to integrate with external tools or services.

3.2.2. Time weighting and indicator scaling

After determining the static indicators, we observe that newer apps in the LLM app store are often disadvantaged in evaluation because they have

Table 2: Static Evaluation Indicator for LLM Apps.

Category	Indicator	Definition
User Engagement	Views	Number of times the app’s main page is viewed by users.
	Interactions	Number of conversions or interactions with the app.
	Favorites	Number of times users add the app to their personal favorites.
	Copies	Number of times users duplicate the app’s configuration.
Functional Capability	Plugins	Number of external services integrated into the app.
	Knowledge Bases	Number of distinct knowledge bases connected to the app.
	Built-in Models	Number of built-in models in the app.

294 had less time to accumulate indicator values, whereas older apps may appear
 295 artificially strong due to the accumulation of historical data. This tempo-
 296 ral accumulation bias can therefore obscure the real-time competitiveness of
 297 apps. To address this issue, we introduce a **time-decay dynamic weight-**
 298 **ing mechanism** that adjusts static indicators according to each app’s opera-
 299 tional age. The core idea is to ensure that the contribution of any cumulative
 300 indicator decreases as time progresses, thereby placing greater emphasis on
 301 recent performance rather than historical accumulation.

302 We first segment the lifecycle of each app into discrete evaluation periods,
 303 using the quarter as the basic analysis unit. Let M denote the total number
 304 of months since an app’s release; the corresponding number of operating
 305 quarters Q is computed as:

$$Q = \max(1, \lfloor M/3 \rfloor) \quad (1)$$

306 where $\lfloor \cdot \rfloor$ represents the floor function, M is the total operation months and
 307 $\max(1, \cdot)$ ensures that even apps with less than three months of operation are
 308 counted as having completed one quarter. This guarantees that both new
 309 and long-standing apps are fairly included in the evaluation.

310 Next, for each of the user engagement indicators (i.e., views, interactions,
 311 favorites, and copies), we sequentially calculate its quarterly average and ap-
 312 ply a time-based decay coefficient. We let X_{total} denote the total cumulative
 313 value of any given user engagement metric recorded over its entire operational
 314 period.

315 The standardized, time-weighted value X_{weighted} is given by:

$$X_{\text{weighted}} = \left(\frac{X_{\text{total}}}{Q} \right) \cdot \omega_t \quad (2)$$

316 This formula first calculates the quarterly mean of the indicator (X_{total}/Q) to
 317 eliminate the total amount difference caused by the operating time. Then, it

reflects the timeliness by multiplying by a decay coefficient (ω_t) that decreases over time. The decay coefficient ω_t is defined as an exponential function of the number of operating quarters Q :

$$\omega_t = \beta^{Q-1} \quad (3)$$

where $\beta \in (0, 1)$ is the decay base that determines the rate at which past performance is discounted. In our implementation, we set $\beta = 0.99$, meaning each additional quarter reduces the weight of historical data to 99% of the previous period. For example, a new app ($Q = 1$) retains full weight ($\omega_t = 1$), while an app active for four quarters ($Q = 4$) is weighted by $\omega_t = 0.99^3 \approx 0.9703$.

This exponential decay mechanism effectively suppresses the inertia of historical data, thereby making the evaluation results more responsive to recent user feedback and app store dynamics. As a result, LAQUAL significantly improves both the timeliness and predictive power of the evaluation process. The decay base β serves as a critical hyperparameter that governs the rate at which historical data are discounted. The choice of β should be tailored to the specific characteristics of the app store, such as platform popularity, the total number of available apps, and the level of user activity. A smaller value of β leads to a faster decay, placing greater emphasis on the most recent market performance. In contrast, a β value closer to one retains more of the historical performance, favoring long-term stability. By fine-tuning β , LAQUAL can be flexibly adapted to different evaluation scenarios and platform characteristics.

3.2.3. Threshold Determination

Building upon the LLM app classification in § 3.1.2 and the time-weighted static indicator evaluation in § 3.2.2, this step focuses on the scientific determination of admission thresholds for each evaluation indicator. Once each LLM app is categorized, we establish a set of **Category-Indicator-Threshold** mapping rules to ensure that the entry criteria are both differentiated by category and grounded in empirical data. Using the AppBuilder as a representative platform, we set baseline thresholds for key indicators across different LLM app categories, as shown in Table 3. For example, *professional* Q&A apps are required to meet a minimum knowledge base size (≥ 1), while *tool-type* apps are primarily assessed by the number of functional components (≥ 2). These thresholds are designed to guarantee that

352 candidate LLM apps possess the essential capabilities, user engagement, and
 353 professionalism expected for their respective categories.

Table 3: Differentiated Admission Thresholds for Different App Categories (AppBuilder).

Indicator Dimension		Professional Q&A	Tool-type	General
User Engagement ¹	Views	≥ 20	≥ 20	≥ 20
	Interactions	≥ 20	≥ 20	≥ 20
	Favorites	≥ 20	≥ 20	≥ 20
	Copies	≥ 20	≥ 20	≥ 20
Functional Capability ²	Plugins	≥ 0	≥ 2	≥ 0
	Knowledge Bases	≥ 1	≥ 0	≥ 0
	Built-in Models	≥ 1	≥ 1	≥ 1

¹ User engagement indicators are weighted by Eq. (2); at least one indicator must meet the threshold.

² Functional capability indicators use original values; all indicators in the corresponding category must meet their thresholds.

354 The thresholds are determined through multiple rounds of empirical val-
 355 idation and data analysis. In particular, user engagement indicators are
 356 closely aligned with the specific characteristics of the LLM app store, such as
 357 platform popularity and user activity levels. Therefore, when adapting this
 358 framework to other platforms, we recommend calibrating these thresholds
 359 based on the actual data distribution of the target environment to ensure
 360 optimal evaluation performance. Once the thresholds are established, we ap-
 361 ply the time-weighted evaluation algorithm described in § 3.2.2 to compute
 362 weighted user engagement scores for each LLM app. Because user engage-
 363 ment indicators such as views and interactions are directly influenced by op-
 364 erating history, LAQUAL uses the time-weighted method defined in Eq. (2)
 365 to reduce the disadvantage faced by newly launched apps. Functional capa-
 366 bility indicators, by contrast, are not affected by operating history and are
 367 therefore evaluated using their original values. During screening, an app is
 368 retained if at least one weighted user engagement indicator meets its thresh-
 369 old, whereas all functional capability indicators in the corresponding category
 370 must satisfy their thresholds.

371 At this stage, we intentionally adopt relatively inclusive thresholds to
 372 efficiently filter out apps with clear functional shortcomings, while minimizing
 373 the risk of excluding promising candidates. This strategy helps maintain
 374 a sufficiently large and high-quality candidate pool for subsequent, more
 375 detailed dynamic evaluations.

3.3. Dynamic Scenario-Adaptive Evaluation

Dynamic scenario-adaptive evaluation is the core of LAQUAL, which addresses the limitations of traditional fixed metrics by dynamically constructing evaluation criteria for a variety of application scenarios. Based on the labels generated for each LLM app by an LLM-based annotator (§ 3.1), the system systematically designs relevant evaluation metrics and scoring criteria. It then generates corresponding evaluation tasks, ensuring that the quantitative evaluation of LLM apps accurately reflects their real-world functions and usage contexts.

3.3.1. Evaluation Metric and Task Generation

For each LLM app, the label generated by the LLM-based annotator (§ 3.1) serves as the anchor for designing targeted evaluation metrics. Guided by the principles of functional uniqueness, quantifiability, and content orientation, LAQUAL constructs a set of metrics that directly correspond to the app’s core capabilities. The prompt template shown in Figure 4 is used to instruct the LLM to generate evaluation metrics based on the given label, resulting in three core metrics for each label. Corresponding scoring criteria are also generated for each evaluation metric. By providing segmented descriptions for each score level, these metrics are translated into clearly structured and quantifiable scoring guidelines.

As a top-tier LLM app quality assessment expert, please design three core content quality assessment metrics for apps labeled "{label}"	**Output format:**
Please ensure that each generated metric and its scoring criteria strictly adhere to all of the following specifications.	Please return in the following JSON format: {{
1. **Basic Requirements** : [.....]	"Metric 1": {{
	"Description": "Metric Description",
	"Rating Criteria": [.....]
	}},
	{{
2. **Core Constraints on Assessment Scope** : [.....]	"Metric 2": {{
In principle, metrics must be based entirely on subjective quality assessments of generated text content. Therefore, any metrics that require external objective data or non-textual content for evaluation are strictly prohibited.	
Specifically, the following aspects are prohibited from being evaluated: [.....]	}},
	{{
	"Metric 3": {{
	
	}}
	}}
3. **Specific Requirements for Scoring Criteria** : [.....]	}}

Figure 4: Prompt Template for Dynamic Evaluation Metric Generation.

Based on these metrics, LAQUAL dynamically generates evaluation tasks tailored to each LLM app. Leveraging the core label and corresponding metrics, LAQUAL simulates real user interaction to create tasks that reflect

399 authentic user needs and diverse usage contexts. LAQUAL can produce de-
 400 tailed and targeted evaluation tasks, with multiple rounds of quality verifica-
 401 tion to ensure both accuracy and reliability. As illustrated in Figure 5, each
 402 evaluation prompt integrates the metric name and description, explicit task
 403 requirements, instructive examples, and relevant content constraints. This
 404 ensures that every task is well-aligned with specific evaluation objectives.

```

  You are a professional LLM app evaluation expert. Please generate a specific, actionable evaluation question based on
  the following information.

  **Evaluation Information**
  * **Evaluation Target:** {metric_name} in the {label} domain
  * **Metric Description:** {description}

  1. **Task Requirements:** Please generate a natural, high-quality user question that strictly meets all of the following
  criteria:<Criteria>
  2. **Reference Examples:**<Examples>
  ** Output Format**

  Please strictly follow the following format and do not include any other content:

  Question: [Question Content]
  
```

Figure 5: Prompt Template for Dynamic Evaluation Tasks Generation.

405 To guarantee task quality, LAQUAL employs a two-stage filtering process.
 406 First, syntactic integrity inspection is performed to eliminate structurally
 407 flawed tasks, such as those with incomplete sentences, incorrect punctuation,
 408 or ambiguous wording. Second, complexity inspection assesses whether the
 409 tasks contain sufficient information, context, and challenge to enable effec-
 410 tive evaluation of the target metric. Tasks that are overly simplistic, vague,
 411 or lacking in necessary detail are filtered out at this stage. If a candidate
 412 task fails either check, a backoff and retry mechanism is triggered: LAQUAL
 413 regenerates the task, with potential adjustments to prompts or constraints,
 414 until all quality criteria are satisfied and a high-quality evaluation task is
 415 produced. The transformation of abstract metrics into concrete, context-
 416 specific evaluation tasks enables LAQUAL to establish a standardized and
 417 adaptable framework. This design supports flexible and comprehensive eval-
 418 uation across a wide range of LLM apps.

419 3.3.2. Response Quality Evaluation.

420 The current response quality evaluation focuses on text-centric LLM apps
 421 whose primary outputs are textual responses. Multimodal LLM apps involv-
 422 ing image, audio, or video generation are not considered in this evaluation.
 423 Building on the evaluation metrics and evaluation tasks generated in § 3.3.1,
 424 we next focus on systematically evaluating the quality of the dynamic inter-
 425 actions between LAQUAL and LLM apps. LAQUAL integrates both **content**
 426 **quality evaluation** and **response performance evaluation**, forming a
 427 closed feedback loop from app evaluation to actionable improvement sugges-
 428 tions. Such a design provides a scientific basis for the functional iteration
 429 and performance tuning of LLM apps.

430 **Content quality evaluation.** This evaluation adopts an LLM-based
 431 evaluation paradigm, in which a structured prompt template guides the eval-
 432 uation model to make quality judgments on LLM app responses. The prompt
 433 template, as illustrated in Figure 6, is constructed using a context embed-
 434 ding approach. It explicitly incorporates three essential elements: *original*
 435 *task*, *response content*, and *scoring criteria*, thus forming a logically com-
 436 plete evaluation context. Semantic alignment between the task and response
 437 is emphasized to ensure the evaluation model accurately understands the cor-
 438 respondence between user intent and generated content. Meanwhile, explicit
 439 scoring criteria transform abstract evaluation metrics into concrete judgment
 440 rules, thereby reducing subjective bias in the evaluation process.

```

You are a professional and objective LLM app evaluation expert. Your
task is to evaluate the quality of the LLM app's answers strictly according
to the scoring criteria provided below, give an integer score between 1 and
5, and provide detailed evaluation reasons and improvement suggestions.

**Information to be evaluated**
* **Original question:** {question}
* **LLM app's answer:** {response}

**Scoring criteria:**
{chr(10).join(scoring_criteria)}

**Evaluation requirements**:
1. **Objective and Fair**
2. **Well-reasoned**
3. **Specific Suggestions**
**Output format:**
Please strictly follow the following JSON format. Do not include any
other content: {{
  "Score": "[An integer between 1 and 5]",
  "Advantages": [.....],
  "Deficiencies": [.....],
  "Improvement suggestions": [.....]
}}
```

Figure 6: Prompt Template for Content Quality Evaluation.

441 To ensure the stability and consistency of evaluation results, model pa-
 442 rameters are finely configured during the LLM invocation. Specifically, we
 443 set `temperature=0` to eliminate randomness and enhance reproducibility;
 444 `frequency_penalty=0.5` to discourage repetitive expressions and increase
 445 informativeness; `max_tokens=5000` to accommodate complex evaluation sce-

446 narios, and `timeout=60` seconds to prevent abnormal response delays. Be-
 447 sides, five-level scale values [1, 5] are used to quantitatively rate content
 448 quality, with each level clearly defined. Alongside the score, the evaluation
 449 model outputs a structured analysis of advantages, weaknesses, and targeted
 450 improvement suggestions. The final quantified score obtained through this
 451 entire systematic evaluation process is defined as the *content quality score*,
 452 denoted as S_{CQ} . This comprehensive framework ensures systematic and ob-
 453 jective evaluation, providing precise guidance for app optimization.

454 **Response performance evaluation.** This evaluation quantitatively
 455 analyzes the real-time interaction capability of LLM apps, with a core focus
 456 on ensuring a fluid user experience. As low latency and high throughput are
 457 critical for smooth real-time interactions, our design centers on a compre-
 458 hensive performance metric: *response efficiency* (η). The metric is defined
 459 as:

$$\eta = \frac{\text{Tokens}}{\text{R_time}} \quad (4)$$

460 where `R_time` denotes the total elapsed time (in seconds) from the user’s
 461 query submission to the receipt of the last token in the LLM-generated re-
 462 sponse, and `Tokens` refers to the total number of tokens in the response. By
 463 collecting real-time performance data during LLM app operation, response
 464 efficiency provides a direct measure of the system’s speed and capability in
 465 processing and generating outputs. To translate this continuous metric into
 466 intuitive and comparable scores, we adopt a discretization strategy based on
 467 predefined efficiency thresholds.

468 Specifically, a lookup table, as shown in Table 4, is designed to map each
 469 measured response efficiency value to a categorical score within [1, 5]. The
 470 final quantified score obtained through this method is defined as the *response*
 471 *performance score*, denoted as S_{RP} . This module offers a quantifiable basis
 472 for performance diagnosis, resource allocation optimization, and service qual-
 473 ity assurance, substantially enhancing the practical and engineering value of
 474 the overall evaluation framework.

Table 4: Response Efficiency Scoring.

Metric Dimension (Quarterly)	Excellent (5 pts)	Good (4 pts)	Average (3 pts)	Poor (2 pts)	Very Poor (1 pt)
Response Efficiency(tokens/sec)	≥ 25	≥ 20	≥ 15	≥ 10	< 10

475 3.3.3. Weighted Composite Scoring

476 After obtaining the independent *content quality score* (S_{CQ}) and *response*
477 *performance score* (S_{RP}), a composite scoring mechanism is applied to ensure
478 that the evaluation results comprehensively and objectively reflect the overall
479 quality of LLM apps. The final composite score is calculated as follows:

$$\text{Score} = \alpha \times S_{CQ} + \lambda \times S_{RP} \quad (5)$$

480 where the weighting coefficients satisfy $\alpha + \lambda = 1$ and $\alpha > \lambda$. Based on
481 empirical analysis, α is set to 0.8 and λ is set to 0.2. The reason is, although
482 response performance is important for user experience, its marginal utility is
483 generally lower than that of content quality. By prioritizing content quality
484 in the weighting, the evaluation system encourages the optimization of sub-
485 stantive output while still considering improvements in response efficiency.
486 The composite evaluation score serves not only as a quantitative indicator of
487 overall application performance, but also as an important basis for iterative
488 optimization. This comprehensive evaluation enables quick identification of
489 both the overall performance level and the specific strengths or weaknesses of
490 an LLM app, providing robust decision support for the continuous improve-
491 ment and targeted tuning of LLM apps.

492 4. Evaluation

493 This section systematically examines the effectiveness of the LAQUAL
494 framework in real-world LLM app scenarios. We focus on evaluation accu-
495 racy, recommendation rationality, and user experience. Our evaluation is
496 guided by the following research questions (RQs):

497 **RQ1: Can LAQUAL provide accurate and scenario-adaptive quality**
498 **evaluation for various types of LLM apps?** This question aims
499 to examine whether the automated evaluation results from LAQUAL
500 are consistent with user judgments and dynamically adapt to different
501 LLM app scenarios.

502 **RQ2: How does LAQUAL compare to recommendation mechanisms**
503 **in mainstream LLM app stores in terms of recommendation**
504 **quality and rationality?** Here, we investigate whether LAQUAL can
505 offer a more comprehensive and reasonable recommendation mecha-
506 nism that helps users identify truly high-quality LLM apps, surpassing
507 existing LLM app store recommendation mechanisms.

508 **RQ3: What is the impact of LAQUAL on user decision support,**
509 **cognitive effort, and trust in real-world LLM app discovery**
510 **tasks?** We focus on determining if the adoption of LAQUAL leads to
511 concrete improvements in user decision-making, reduces mental work-
512 load, and enhances trust during the LLM app selection process.

513 4.1. RQ1: Accuracy and Adaptivity

514 4.1.1. Experimental Setup

515 All experiments were conducted on the AppBuilder [4], a mainstream
516 LLM app store, which provides a diverse set of LLM apps and a rich array
517 of functional configurations. Two distinct scenarios were selected as experi-
518 mental subjects to ensure coverage of both *professional Q&A* and *tool-type*
519 LLM app categories: (1) *legal consulting* and (2) *travel planning*. For each
520 scenario, a single scenario-specific keyword was used to retrieve candidate
521 LLM apps from the platform, reflecting the typical user search and recom-
522 mendation logic in LLM app stores. Specifically, the keyword “lawyer” was
523 used for the *legal consulting* scenario, which yielded 18 candidate apps, while
524 “smart travel planning assistant” was used for the *travel planning* scenario,
525 yielding 32 candidate apps. These numbers represent the original set of apps
526 to be evaluated for each scenario.

527 Subsequently, each set of candidate apps was independently evaluated
528 by LAQUAL. For all LLM-based evaluation modules in LAQUAL, we used
529 “Qwen-32B”. It is worth noting that the underlying LLM in LAQUAL is mod-
530 ular and can be replaced with other LLMs as needed. In parallel, a panel
531 of 28 human evaluators, including several domain experts, assessed the same
532 LLM apps using the same metrics. All evaluators were experienced LLM app
533 users aged 18–50, with several possessing relevant domain expertise.

534 4.1.2. Automated Evaluation by LAQUAL

535 The automated evaluation process in LAQUAL comprises three main
536 stages: (1) LLM app labeling and classification, (2) static indicator eval-
537 uation, and (3) dynamic scenario-adaptive evaluation.

538 **LLM app labeling and classification.** Before formal evaluation, LAQ-
539 UAL employs an LLM-based annotator to automatically generate concise,
540 scenario-specific labels for each app based on its name and description. These
541 labels distill the core functionality and typical usage scenario of the app,
542 facilitating accurate downstream classification. Among the 18 LLM apps

retrieved using the keyword “lawyer” in the legal consulting scenario, the annotator consistently generated the label “law consulting analysis” for these apps. According to the functional taxonomy, this label corresponds to the “Law” tag, which is categorized under the *professional* Q&A category. Similarly, for the 32 LLM apps retrieved with the keyword “smart travel planning assistant” in the travel planning scenario, the annotator generated the label “travel itinerary planning”. This label maps to the “Planning” tag, which falls under the *tool-type* category in the taxonomy. Each app is thus mapped to a fine-grained hierarchical taxonomy, including category, subcategory, and tag. This process ensures that subsequent evaluation criteria and thresholds are precisely aligned with the app’s intended usage context, thereby enabling scenario-adaptive evaluation.

Static indicator evaluation. Static evaluation aims to efficiently reduce the candidate pool size and improve baseline app quality before dynamic evaluation. For the legal consulting scenario, the initial set of 18 apps was filtered using a combination of user engagement metrics (views, interactions, favorites, copies) and functional capability metrics (plugins, knowledge bases, built-in models), together with category-specific thresholds and time-decay weighting. As a result, LAQUAL selected 6 high-potential candidates. Similarly, in the travel planning scenario, the candidate pool was reduced from 32 to 6 apps. Table 5 summarizes the original static indicator values for all shortlisted apps. During screening, user engagement indicators were evaluated using weighted values, whereas functional capability indicators were compared using their original values. All retained apps satisfied the screening requirements, ensuring a suitable candidate pool for subsequent quality evaluation.

Table 5: Static Indicator Values for Shortlisted LLM Apps.

Category	Metric	La1 ¹	La2	La3	La4	La5	La6	Ta1 ²	Ta2	Ta3	Ta4	Ta5	Ta6
User Engagement	Views	154	26	19	28	271	70	20	103	17	31	12	14
	Interactions	425	157	78	234	15,000	68	132	59	117	345	169	36
	Favorites	1	0	0	0	4	0	2	0	0	0	0	0
	Copies	0	1	0	0	0	0	0	1	0	1	1	0
Functional Capability	Plugins	3	0	4	2	0	6	5	3	4	6	4	4
	Knowledge Bases	15	1	1	1	2	1	2	1	1	1	1	1
	Built-in Models	2	2	2	2	2	2	2	2	2	2	2	2

¹ LaX denotes apps in the *legal consulting* scenario (La = Legal consulting app).

² TaX denotes apps in the *travel planning* scenario (Ta = Travel planning app).

Dynamic scenario-adaptive evaluation. Following static screening, the dynamic evaluation module of LAQUAL automatically adapts its met-

rics to the application’s scenario, leveraging the previously generated labels. For every shortlisted app, the framework dynamically constructs three core evaluation metrics tailored to the scenario, and for each metric, it defines a five-level quantitative scoring metric, values set as [1, 5], with detailed criteria for each level. Representative scenario-specific evaluation tasks are designed to ensure the assessment’s authenticity and relevance. For the legal consulting and travel planning scenario, LAQUAL establishes three core evaluation metrics for each context, as presented in Table 6. The scoring metrics ensure that different levels of app performance are clearly and fairly distinguished.

Table 6: Scenario-Adaptive Dynamic Evaluation Metrics and Tasks in LAQUAL.

Scenario	Metric	Definition	Task Example	Score
Legal Consulting	Legal Citation Accuracy	Assesses whether legal terms cited in the answer are accurate and relevant to the question.	“If an employee is dismissed without notice, which labor law articles apply?”	1–5
	Terminology Clarity	Evaluates whether legal terminology is used appropriately and clearly explained.	“Explain the meaning of ‘force majeure’ in a contract dispute to a layperson.”	1–5
	Answer Completeness & Logic	Judges if the answer addresses key legal points, is logically organized, and has a well-founded conclusion.	“I’m a tenant and I need to break my lease early. What are the legal steps I need to follow? What risks should I be aware of?”	1–5
Travel Planning	Personalization Match	Measures whether the plan meets all personalized requirements (e.g., budget, interests, time constraints).	“Plan a 3-day trip to Chengdu for a family with kids, with a budget under 3000 RMB.”	1–5
	Itinerary Logic & Flow	Assesses the reasonableness and flow of the itinerary, including transportation and activity planning.	“Design a 5-day Japan itinerary covering Tokyo, Kyoto, and Osaka with minimal backtracking.”	1–5
	Content Innovation	Evaluates whether recommendations include unique and insightful local attractions or activities.	“Recommend lesser-known attractions or local experiences in Florence for art lovers.”	1–5

LLM apps demonstrated varying performance across core metrics. In the legal consulting scenario, most apps received scores of 3–4, indicating generally good performance, while a few had lower scores on certain metrics, such as legal citation accuracy. In the travel planning scenario, top apps consistently scored higher, while others showed weaknesses in itinerary logic and personalization. These results highlight the framework’s ability to provide fine-grained, scenario-relevant evaluation and to identify both strengths and weaknesses of each app.

4.1.3. Human Evaluation and Consistency Analysis

To benchmark the accuracy and adaptivity of LAQUAL, the six short-listed apps from each scenario were independently evaluated by the panel of 28 human raters, including domain experts, using the same dynamically generated metrics and a 5-point scale. We then compared the scores from LAQUAL with the average human ratings.

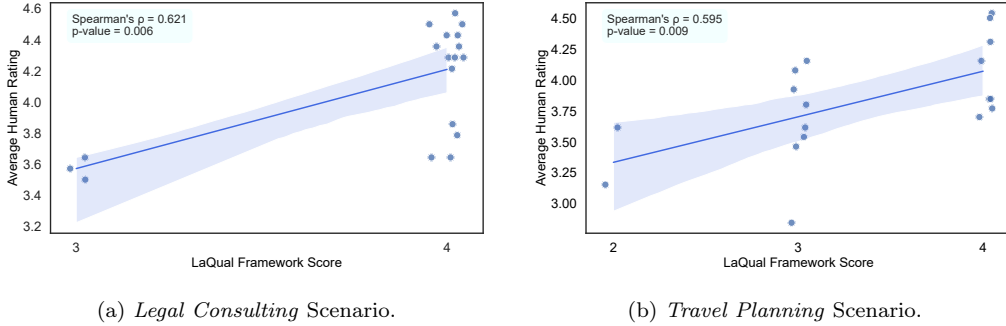


Figure 7: Correlation Analysis between LAQUAL and Human Evaluation.

For the legal consulting scenario, Spearman’s rank correlation analysis was conducted between the human average scores and LAQUAL’s content scores across all 18 evaluation items (six apps, three metrics each). As shown in Figure 7a, the correlation coefficient (ρ) [35] was **0.621** with a p-value of **0.006**, indicating a moderately strong and statistically significant positive correlation. This demonstrates that LAQUAL’s automated evaluation is highly consistent with human expert judgment. Regarding response performance, it’s noted that no direct correlation with human ratings was assessed due to differences in measurement: human scores reflect subjective fluency perceptions, while LAQUAL computes objective response efficiency. Nevertheless, the response performance score (S_{RP}) remains a critical component of the final composite evaluation, ensuring that both content quality and interaction fluency are reflected in the overall assessment.

Similarly, for the travel planning scenario, the consistency between LAQUAL and human raters was evaluated using the same method. As shown in Figure 7b, Spearman’s correlation coefficient was **0.595** with a p-value of **0.009**, again indicating a moderately strong and statistically significant positive correlation. This further validates LAQUAL’s adaptability and accuracy across different application domains.

Answer to RQ1: The results show that LAQUAL achieves high consistency with human judgments across diverse LLM app scenarios. In the legal consulting case, the Spearman correlation coefficient between LAQUAL’s automated scores and user ratings reached 0.621 ($p = 0.006$); in the travel planning case, the coefficient was 0.595 ($p = 0.009$). This indicates that LAQUAL not only provides accurate evaluations, but also dynamically adapts its assessment criteria to fit different domains.

614 4.2. RQ2: Recommendation Effectiveness

615 RQ2 conducts a further analysis based on the experimental results of RQ1
616 to examine whether LAQUAL can provide more effective and rational recom-
617 mendations than mainstream LLM app store recommendation mechanisms.

618 **Manual verification procedure and consensus.** To better under-
619 stand the exclusion patterns produced by the static screening stage, we con-
620 ducted a qualitative verification of the 38 excluded apps based on consensus
621 among the volunteers (12 in the legal scenario and 26 in the travel plan-
622 ning scenario). This verification was carried out by three volunteers with
623 relevant experience in software quality research. Specifically, the verification
624 was conducted in two phases: (1) *Independent Examination*, in which each
625 volunteer independently examined the apps’ metadata, platform indicators,
626 and functional configurations; and (2) *Consensus Discussion*, in which the
627 team collectively discussed the results to reach agreement on the major exclu-
628 sion patterns. The verification suggests that the excluded apps consistently
629 exhibited the following recurring characteristics:

- 630 • **Lack of market vitality:** zombie apps with very low or declining user
631 activity, or early-stage apps that had not yet gained meaningful user recog-
632 nition.
- 633 • **Lack of domain support:** apps that relied mainly on foundation model
634 knowledge and lacked integrated knowledge bases or other support relevant
635 to the scenario.
- 636 • **Functional deficiencies:** especially in the travel planning scenario, apps
637 that lacked essential real-time tool integrations (e.g., map navigation and
638 weather plugins).
- 639 • **Superficial design:** apps whose functions or interaction design showed
640 limited differentiation and failed to address the core needs of the target
641 scenario.

642 **Comparison with the default app store ranking.** We further com-
 643 pared LAQUAL’s selected apps with the store’s default *top-6 most popular*
 644 results. We observed a partial mismatch between the store’s popularity based
 645 ranking and LAQUAL’s screening results: LAQUAL retained only 3 of the
 646 store’s Top-6 apps in the legal scenario and 2 of the store’s Top-6 apps in
 647 the travel planning scenario. Among the highly ranked platform apps that
 648 were still filtered out, we observed recurring limitations in support relevant
 649 to the scenario. In particular, a common pattern was the lack of plugins or
 650 knowledge base support needed for the scenario. In some cases, these apps
 651 also exhibited imbalanced engagement signals, such as single-digit page views
 652 paired with hundreds or even thousands of interactions. By contrast, several
 653 apps retained by LAQUAL but not ranked in the platform’s default Top-
 654 6 showed more complete functional configurations than some of the highly
 655 ranked apps that were filtered out. These results suggest that LAQUAL does
 656 not simply reproduce popularity-based ranking, but helps identify apps that
 657 are more consistent with scenario-specific requirements.

Answer to RQ2: LAQUAL can provide more effective and rational recom-
 mendations than mainstream LLM app store recommendation mechanisms.
 For example, in the legal consulting scenario, LAQUAL reduced the candi-
 date pool from 18 to 6 apps (a 66.7% reduction), and in the travel planning
 scenario, from 32 to 6 apps (an 81.25% reduction), by systematically filtering
 out apps that did not satisfy the user engagement or functional requirements
 of the framework. Manual verification further suggests that the excluded
 apps typically lacked sufficient market vitality, domain support, or essential
 functional capabilities, indicating that LAQUAL captures recommendation
 cues beyond popularity alone and helps identify apps that are better aligned
 with the requirements of the scenario.

658 4.3. RQ3: User Impact

659 This RQ examines the practical value of the framework in real-world LLM
 660 app discovery scenarios. Specifically, it investigates whether the dynamic
 661 evaluation mechanism provided by LAQUAL can substantially improve deci-
 662 sion support, cognitive efficiency, and trust, compared to a baseline system.
 663 The overarching goal is to assess the framework’s effectiveness as a decision
 664 support tool for LLM app selection.

665 4.3.1. *Research Hypotheses*

666 Based on the core challenges users face in the LLM app store, such as in-
667 formation overload and difficulty discerning quality, we propose the following
668 hypotheses:

- 669 • **RH1: Decision support advantage.** One system may better help users
670 identify apps that match their core needs, leading to more confident final
671 choices.
- 672 • **RH2: Cognitive efficiency advantage.** One system may reduce users’
673 mental effort during app comparison and selection, thereby improving
674 comparison efficiency and decision confidence.
- 675 • **RH3: Trust and transparency advantage.** One system may be per-
676 ceived as more transparent, credible, and helpful for supporting app dis-
677 covery decisions.

678 4.3.2. *Task Design*

679 To empirically validate these hypotheses, we conducted a controlled user
680 study with a within-subjects design. Twelve participants were recruited to
681 use both the LAQUAL framework and a baseline system (i.e., AppBuilder)
682 for app discovery and selection under the same application scenario and de-
683 cision goal. The study focused on two application scenarios related to legal
684 consulting and travel planning. To mitigate order effects, participants were
685 randomly assigned to two groups in a counterbalanced sequence:

686 **Group A (n=6):** Used the baseline system first, then LAQUAL.

687 **Group B (n=6):** Used LAQUAL first, then the baseline system.

688 For each scenario-based app discovery task, participants entered scenario-
689 relevant keywords in both conditions. Examples included “lawyer” and “civil
690 law” for the legal scenario, and “travel itinerary planning” and “travel guide”
691 for the travel scenario. Participants were allowed to choose their own key-
692 words within each predefined scenario, as users may have different needs
693 even within the same broad scenario. In the baseline condition, participants
694 entered the keyword into the live AppBuilder platform, browsed the apps re-
695 turned by the default search and ranking mechanism, and identified suitable
696 apps. The displayed information included app names, descriptions, rankings,

697 and engagement information provided by the platform. In the LAQUAL con-
 698 dition, participants entered the same keyword into the LAQUAL console, re-
 699 viewed the apps recommended by LAQUAL, and made their selections based
 700 on the generated evaluation report. For each recommended app, LAQUAL
 701 provided a direct trial URL, the overall evaluation score, scenario-specific
 702 metric scores, and the corresponding advantages, deficiencies, and improve-
 703 ment suggestions for each evaluation metric.

704 Although the exact keywords varied across participants, each participant
 705 used the same keyword in both conditions for the same task. This within-
 706 subject design allowed the two systems to be compared under the same user
 707 need for each participant. The study was conducted through direct inter-
 708 action with the live platform and LAQUAL. Figure 8 shows representative
 709 interface excerpts from the two conditions: the baseline search-result inter-
 710 face and the recommendation report generated by LAQUAL after the same
 711 keyword was entered.

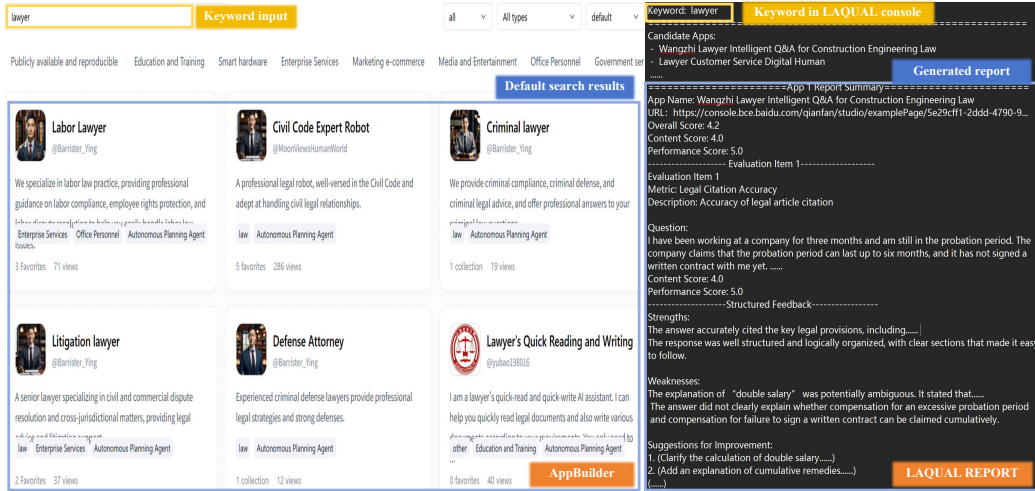


Figure 8: Representative excerpts from the baseline AppBuilder results and the recommendation report generated by LAQUAL.

712 After completing all tasks, participants filled out a post-task questionnaire
 713 based on a seven-point Likert scale (1 = Strongly Disagree, 7 = Strongly
 714 Agree). The questionnaire was designed around the three research hypotheses
 715 and covered six evaluation dimensions, as detailed in Table 7.

Table 7: Quantitative Evaluation Dimensions and Metrics for the User Study.

Dimension	Weight	Metric	Description
Decision Support Benefits	15%	Decision Confidence	The information provided by the system ¹ helped me make a final choice with confidence.
	15%	Comparison Efficiency	The system helped me compare different LLM apps more efficiently, reducing my cognitive and decision-making load.
Evaluation Quality & Credibility	15%	Value of Explanatory Information	The information and evidence provided by the system were substantively helpful for my decision-making.
	15%	Result Credibility	I found the results provided by the system objective and reliable.
Depth of Demand Matching	20%	Deep Demand Fulfillment	The system provided results that addressed both my immediate needs and relevant considerations.
Exploration & Discovery Value	20%	Discovery of “Surprise”	The system helped me identify additional app options worth considering.

¹ “The system” refers to either LAQUAL or Baidu Qianfan.

4.3.3. Results

We analyzed the questionnaire data using a paired-samples t-test, as each participant experienced both systems. The results, summarized in Table 8, indicate that LAQUAL performed better than the baseline system across all reported evaluation dimensions.

As visualized in Figure 9, LAQUAL achieved notably higher mean scores (all statistically significant at the $p < 0.001$ level, as indicated by *** in the figure). The advantage was particularly pronounced in the dimensions of *comparison efficiency* (M=5.45 vs. M=3.30) and *Value of Explanatory Information* (M=4.75 vs. M=2.25). These results suggest that the information provided by LAQUAL helps users make app comparisons more efficiently and confidently.

Furthermore, LAQUAL also showed higher scores in *decision confidence*, *result credibility*, and the *discovery of “surprise”*, further indicating its positive effects on user confidence and trust.

Table 8: Paired-Samples T-Test Results Comparing LAQUAL and the Baseline System.

Evaluation Category	Sub-dimension	LAQUAL Mean	Baseline Mean	LAQUAL SD	Baseline SD
Decision Support Benefits	Decision Confidence (Dim. 1)	5.050	4.400	0.686	0.503
	Comparison Efficiency (Dim. 2)	5.450	3.300	0.887	0.733
Evaluation Quality & Credibility	Value of Explanatory Information (Dim. 3)	4.750	2.250	0.716	0.716
	Result Credibility (Dim. 4)	4.850	3.900	0.587	0.852
Depth of Demand Matching	Deep Demand Fulfillment (Dim. 5)	4.300	3.700	0.733	0.571
Exploration & Discovery Value	Discovery of “Surprise” (Dim. 6)	4.950	3.550	0.686	0.510
Weighted Total Score	Overall Experience Score	4.873	3.555	-	-

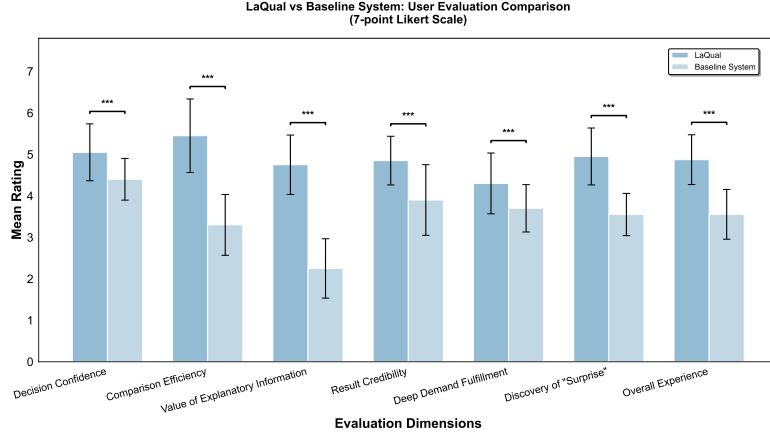


Figure 9: Mean Score Comparison of LAQUAL and the Baseline System Across All User Evaluation Dimensions.

Answer to RQ3: User studies indicate that LAQUAL improves users’ decision support, reduces cognitive effort, and enhances trust. Quantitative results show that LAQUAL achieved higher mean scores than the baseline system across all reported evaluation dimensions, with particularly strong advantages in comparison efficiency (M=5.45 vs. M=3.30) and value of explanatory information (M=4.75 vs. M=2.25). These findings suggest that LAQUAL contributes to more confident, efficient, and trustworthy LLM app selection experiences.

5. Implications

Our evaluation framework provides actionable benefits and concrete guidance for all key stakeholders in the LLM app ecosystem, supporting the establishment of a more transparent, high-quality, and trustworthy environment.

LLM app store managers. For store managers, our findings demonstrate that relying solely on static metrics such as downloads or user ratings can easily result in ranking manipulation and poor user experience. By integrating LAQUAL into routine platform operations, managers can implement automated quality evaluations at both the onboarding and periodic review stages. For example, only apps that meet minimum quality thresholds in scenario-adaptive evaluation can be listed, and regular re-evaluations can

742 help identify apps with declining performance for timely intervention or re-
743 moval. Additionally, publishing composite and detailed evaluation scores
744 alongside traditional popularity indicators allows users to make more in-
745 formed decisions, while also encouraging developers to focus on genuine qual-
746 ity improvements.

747 **LLM app developers.** For developers, LAQUAL offers a set of action-
748 able, scenario-specific quality metrics that can be integrated directly into
749 development and release workflows. Developers can use these evaluations
750 during prototyping to quickly identify weaknesses in content, functionality, or
751 user experience, and address them before public release. Incorporating LAQ-
752 UAL into continuous integration pipelines enables automatic quality checks
753 with each update, reducing the risk of quality regressions. Furthermore, de-
754 tailed evaluation reports can help developers prioritize improvements in areas
755 that matter most to users, such as response accuracy or feature completeness.
756 By transparently sharing evaluation results, developers can also strengthen
757 user trust and better position high-quality LLM apps in a competitive mar-
758 ket.

759 **LLM app users.** End users benefit from our framework through greater
760 transparency and confidence in the LLM apps they select. With access to
761 multi-dimensional evaluation results, users can choose apps that best match
762 their individual needs. For instance, prioritizing response accuracy in legal
763 apps or customization in travel apps. The availability of sub-scores and
764 scenario-based explanations also makes it easier for users to understand the
765 strengths and weaknesses of each app, reducing decision fatigue and the risk
766 of disappointment. As a result, users are empowered to make more informed
767 and satisfying choices, and can provide valuable feedback to further refine
768 quality standards.

769 **Researchers.** For the research community, this research contributes two
770 core concepts to the field of LLM app quality evaluation. The first is the
771 adaptability of the evaluation model. We revealed that the inherent judg-
772 ment logic of the LAQUAL evaluation framework varies depending on the
773 attributes of the metric. Although all evaluations ultimately output a unified
774 five-point scale, the framework’s internal logic for deriving this score shifts
775 based on the inherent differences in the app scenario. When evaluating met-
776 rics with clear objective criteria or knowledge boundaries, such as “accuracy
777 of legal references” the criteria are highly quasi-binary, making qualitative
778 grading easier. However, when evaluating metrics that essentially measure
779 “completeness” or “match” such as “personalized match of travel plans” the

780 focus is more on the completeness of the multi-functional package or per-
781 sonalized satisfaction, resulting in a wider distribution of final scores. We
782 believe that it is precisely through this flexibility in its inherent judgment
783 logic that LAQUAL achieves accurate evaluation across scenarios.

784 Secondly, we reveal that tool acts not just as an evaluator, but also as a
785 **stabilizer** for the evaluation ecosystem. It should be considered a stabilizer
786 within the evaluation ecosystem. A recurring phenomenon in our empirical
787 research is that the ratings from human evaluation panels often exhibit sub-
788 stantial dispersion, reflecting significant differences between evaluators due
789 to subjective experience and preferences. In contrast, LAQUAL provides an
790 objective and stable rating for app evaluations, acting as a stabilizer.

791 6. Threats to Validity

792 Despite the comprehensive and systematic design of the LAQUAL frame-
793 work, several threats to validity remain. We recognize these limitations and
794 have taken targeted measures to mitigate their impact where possible. This
795 section discusses threats from the perspectives of content, internal, and ex-
796 ternal validity.

797 **Content validity.** In the static indicator evaluation, content validity
798 may be affected by the subjectivity of threshold setting. Since complete
799 platform data is unavailable, the admission threshold in this framework is
800 determined by practical experience, and its optimality requires further valida-
801 tion. Additionally, fixed thresholds may inadvertently exclude apps targeting
802 niche scenarios. To minimize these threats, we adopt a low-threshold filter-
803 ing strategy to first eliminate unqualified applications, reducing the risk of
804 filtering out potential high-quality apps. We also set differentiated functional
805 indicator requirements for different app categories. While this approach is
806 preliminary, it considers the characteristics of various apps and maintains
807 filtering efficiency, enabling the framework to achieve scenario-based and ac-
808 curate evaluation while retaining overall generality.

809 **Internal validity.** The internal validity of our evaluation may be affected
810 by inherent bias within LLMs. Since the evaluation process is powered by
811 a LLM, it is susceptible to political, cultural, or social biases present in its
812 training data or architecture [36, 37]. These biases can influence scoring
813 results, particularly in subjective or open-ended scenarios, and may compro-
814 mise fairness and objectivity. To mitigate this risk, several strategies are
815 employed. First, we use carefully designed prompt instructions to constrain

816 LLM behavior. Second, we adopt standardized and scenario-adaptive scoring
817 metrics to minimize ambiguity. Third, we set deterministic decoding param-
818 eters, such as temperature at zero, to enhance the consistency of results.
819 Despite these measures, it is not possible to fully eliminate model-internal
820 bias, and interpretation of evaluation outcomes should take this limitation
821 into account.

822 **External validity.**

823 Several factors may affect the generalizability of our findings. The re-
824 ported evaluation covered two scenarios: 18 LLM apps for legal consulting
825 and 32 LLM apps for travel planning. These scenarios include both pro-
826 fessional Q&A and tool-type apps, and the results should be interpreted as
827 evidence from a focused empirical study. The evaluation was conducted on
828 Baidu AppBuilder. Because LLM app stores may differ in ranking mecha-
829 nisms, metadata structures, and user engagement indicators, some compo-
830 nents of LAQUAL, such as static thresholds, may need to be calibrated before
831 the framework is used on other platforms. In addition, the evaluation used
832 concise search terms to reflect common app discovery behavior. Future work
833 should consider more diverse and complex query settings to further test the
834 robustness of the framework. The experiments were conducted in the Chi-
835 nese platform environment of Baidu AppBuilder, using Chinese prompts and
836 app content during the actual evaluation. For readability and consistency,
837 the examples in this paper are presented in English. The framework itself
838 is not tied to a specific language; when LAQUAL is applied to platforms in
839 other linguistic environments, the prompts, taxonomy terms, and scenario
840 descriptions can be adapted to the target language.

841 In addition, the current LAQUAL is limited to evaluating text-based out-
842 puts of LLM apps. As the LLM app ecosystem is evolving toward multi-
843 modal applications, including those involving image generation and audio-
844 visual content, our evaluation methods may not be directly applicable to
845 these new forms. To partially mitigate these limitations, we designed our
846 evaluation tasks and metrics to be scenario-adaptive and structurally exten-
847 sible, providing a foundation for future adaptation to multi-modal or hybrid
848 applications. However, the current conclusions should be interpreted within
849 the scope of text-centric LLM app scenarios, and extending the framework to
850 multi-modal evaluation remains an important direction for future research.

851 7. Related Work

852 7.1. Research on LLM Apps

853 The explosive growth of LLM apps has given rise to a novel software
854 ecosystem centered on LLM app stores. Although academic research on this
855 ecosystem is still in its infancy, significant strides have already been made in
856 several key areas. Current studies primarily concentrate on three dimensions:
857 ecosystem analysis, security vulnerabilities, and data privacy.

858 At the ecosystem level, pioneering efforts have begun to map the land-
859 scape of this rapidly developing field. Zhao et al. [1] laid the groundwork by
860 proposing the first comprehensive research roadmap for the LLM app ecosys-
861 tem. Foundational datasets and initial exploratory analyses were contributed
862 by Hou et al. [38], who constructed the large-scale GPTZoo dataset, and by
863 Yan et al. [8], who investigated patterns in LLM app distribution. In terms
864 of security, Hou et al. [11] were among the first to expose the security threats
865 facing LLM app stores, systematically identifying several attack vectors such
866 as app hijacking and prompt injection. Building on this, Xie et al. [39] con-
867 ducted an in-depth analysis of app squatting and cloning, highlighting the
868 severe risks these imitation attacks pose to the overall trust and stability of
869 the ecosystem. Regarding data privacy, researchers have revealed that the
870 flexibility inherent in app configuration can also introduce significant risks of
871 data leakage. Li et al. [9] demonstrated that private application knowledge
872 files are susceptible to theft via prompt engineering, while Wu et al. [10]
873 uncovered the widespread issue of excessive user data collection throughout
874 the ecosystem. While existing research has made notable advances in under-
875 standing the ecosystem’s structure as well as its security and privacy risks,
876 most efforts remain focused on external threats and vulnerabilities. Little
877 attention has been paid to systematically evaluating the intrinsic quality of
878 content generated by LLM apps. This represents an important research gap.
879 The framework proposed in this study, LAQUAL, seeks to fill this gap by
880 enabling systematic and automated evaluation of LLM app content quality.

881 7.2. Domain-Specific LLM Evaluation

882 A growing body of research has focused on evaluating LLMs within spe-
883 cialized domains, resulting in a variety of tailored frameworks and bench-
884 marks that address the unique requirements and challenges of different pro-
885 fessional fields. In the healthcare domain, recent work has addressed both
886 the opportunities and challenges of applying LLMs to medical tasks. Krolik

et al. [40] explored the use of LLMs for automated medical Q&A, while Nazi et al. [41] argued that evaluation should consider not only generative ability but also practical applicability and safety in real-world scenarios. Building on this, Ei et al. [42] introduced a dedicated evaluation framework for medical queries, finding that ChatGPT achieves high accuracy in this context. In finance, Guo et al. [43] proposed FinLMEval to systematically evaluate LLMs on a range of financial tasks. Xie et al. [44] further developed FinBen, the first comprehensive benchmark for financial LLM evaluation, enabling more detailed assessment of model performance in complex financial domains. In education, Xiao et al. [45] built a system for generating reading comprehension exercises using LLMs, validating its effectiveness through both automatic and manual evaluation. Kavadella et al. [46] examined ChatGPT’s role in undergraduate dental education, showing improved student performance with model assistance. The legal field has also seen robust evaluation efforts. Wang et al. [47] compared LLM-generated legal content to reference materials and expert judgments. Li et al. [48] presented LegalAgentBench, which uses metrics such as success rate, progress rate, and BERT-Score for hierarchical legal tasks. Haitao Li et al. [49] introduced LexEval for systematic evaluation in the Chinese legal context, while Savelka et al. [50] showed that GPT-3.5 achieves high accuracy in zero-shot legal text annotation.

These domain-specific evaluation frameworks underscore the substantial promise of LLMs in professional and expert-driven contexts. However, their scope is typically confined to individual scenarios, lacking the scalability and adaptability required for broader application landscapes. This limitation highlights a central motivation for our work: to move beyond fixed-scenario evaluations and develop a universal, automated framework capable of dynamically generating scenario-aware evaluation criteria for LLM apps.

7.3. Metrics for LLM evaluation

Mainstream evaluation of LLM content quality largely relies on traditional natural language processing metrics. Surface-level semantic matching is typically assessed using BLEU [51] and ROUGE [52], which measure text similarity based on n-gram statistics such as word order and vocabulary overlap. METEOR [53] improves upon these by enhancing the ability to capture semantic equivalence between texts. More recently, metrics like BERTScore [54] and BLEURT [55] have advanced the field by leveraging pre-trained language models to assess deep semantic similarity, representing a shift from pure surface matching to semantic-level evaluation. However,

924 these methods still exhibit notable limitations in the context of LLM eval-
925 uation. Specifically, they often lack domain adaptation and deep semantic
926 understanding, making it difficult to accurately assess outputs that involve
927 creative expression or specialized domain knowledge. As a result, there can
928 be significant discrepancies between automated evaluation scores and the
929 actual semantic quality perceived by users in real-world LLM applications.

930 7.4. *Evaluation Based on Human Preferences*

931 While traditional algorithm-based evaluation methods offer efficiency,
932 they often struggle to capture the nuanced preferences and subjective judg-
933 ments of human users. Belz and Reiter [56] provided a comprehensive review
934 of human evaluation techniques for natural language generation, underscor-
935 ing the indispensable value of human judgment in assessing generated text
936 quality. To better align models with user expectations, Ouyang et al. [57] in-
937 troduced reinforcement learning from human feedback (RLHF), where mod-
938 els are trained using human preference rankings for different outputs. Simi-
939 larly, Chatbot Arena [58] employs crowdsourcing to evaluate LLMs based on
940 large-scale human preferences, while Shankar et al. [59] proposed EvalGen,
941 a hybrid framework aiming to better align evaluators with human expecta-
942 tions. In the context of code generation, Yang et al. [31] highlighted the
943 critical importance of model alignment with human preferences through sys-
944 tematic evaluation. Although these approaches mark a shift from traditional
945 semantic matching to preference-based assessment, they generally require
946 extensive human annotation and face challenges in maintaining annotation
947 consistency. Consequently, their scalability and practicality for automated,
948 large-scale evaluation in LLM app ecosystems remain limited.

949 8. Conclusion

950 In this work, we present LAQUAL, a unified and automated framework for
951 large-scale LLM app quality evaluation. Our approach systematically inte-
952 grates scenario-aware hierarchical classification, multi-dimensional static in-
953 dicators filtering, and LLM-driven dynamic evaluation to provide fine-grained
954 and context-adaptive assessment across diverse LLM app scenarios. Exten-
955 sive experiments on real-world app store data demonstrate that LAQUAL
956 achieves strong alignment with expert judgments, substantially reduces the
957 candidate app pool, and significantly improves user decision quality, effi-
958 ciency, and trust compared to existing recommendation mechanisms. Fur-

thermore, our framework is highly extensible and can be adapted to new evaluation scenarios as the LLM app ecosystem evolves. Moving forward, we plan to further enhance the interpretability of automated evaluation results and explore integration with real-time user feedback, aiming to further enhance the robustness and trustworthiness of LLM app discovery experiences.

CRedit authorship contribution statement

Yan Wang: Writing - original draft, Writing - review & editing, Software, Investigation, Formal analysis, Methodology, Visualization.

Xinyi Hou: Writing - review & editing, Methodology, Validation, Investigation, Visualization.

Junjun Si: Writing - review & editing, Supervision, Methodology, Investigation.

Yanjie Zhao: Writing - review & editing, Methodology, Validation.

Weiguo Lin: Writing - review & editing, Methodology, Supervision.

Haoyu Wang: Writing - review & editing, Supervision, Resources, Methodology.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT in order to improve the English language clarity and grammar of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data Availability

The implementation details and evaluation results for LAQUAL are publicly available at <https://github.com/wangyanainn-ainn/LaQual>.

Acknowledgements

The authors would like to thank our colleagues and peers at our respective institutions for their constant support and insightful discussions. We also sincerely appreciate the volunteers and domain experts who participated in the user studies and consistency assessments. This work was sponsored by the

989 State Key Laboratory of Multimedia Information Processing Open Fund and
990 the Fundamental Research Funds for the Central Universities under Grant
991 CUC25XT02.

992 References

- 993 [1] Y. Zhao, X. Hou, S. Wang, H. Wang, Llm app store analysis: A vision
994 and roadmap, *ACM Trans. Softw. Eng. Methodol.* 34 (5) (May 2025).
995 doi:10.1145/3708530.
996 URL <https://doi.org/10.1145/3708530>
- 997 [2] OpenAI, ChatGPT, <https://chatgpt.com/gpts> (2023).
- 998 [3] Quora, Poe, <https://poe.com> (2024).
- 999 [4] Baidu, Baidu app center, [https://console.bce.baidu.com/ai_apaas](https://console.bce.baidu.com/ai_apaas/appCenter)
1000 /appCenter (2024).
- 1001 [5] Tencent, Tencent agent shop, [https://yuanqi.tencent.com/agent-s](https://yuanqi.tencent.com/agent-shop)
1002 hop (2024).
- 1003 [6] ByteDance, Coze, <https://www.coze.com/> (2024).
- 1004 [7] ByteDance, Cici, <https://www.cici.ai.com/chat/> (2024).
- 1005 [8] C. Yan, R. Ren, M. H. Meng, L. Wan, T. Y. Ooi, G. Bai, Explor-
1006 ing chatgpt app ecosystem: Distribution, deployment and security, in:
1007 Proceedings of the 39th IEEE/ACM International Conference on Au-
1008 tomated Software Engineering, ASE '24, Association for Computing
1009 Machinery, New York, NY, USA, 2024, p. 1370–1382. doi:10.1145/
1010 3691620.3695510.
1011 URL <https://doi.org/10.1145/3691620.3695510>
- 1012 [9] C. Yan, B. Guan, Y. Li, M. H. Meng, L. Wan, G. Bai, Understanding
1013 and detecting file knowledge leakage in gpt app ecosystem, in: Pro-
1014 ceedings of the ACM on Web Conference 2025, WWW '25, Association
1015 for Computing Machinery, New York, NY, USA, 2025, p. 3831–3839.
1016 doi:10.1145/3696410.3714755.
1017 URL <https://doi.org/10.1145/3696410.3714755>

- 1018 [10] Y. Wu, E. Jaff, K. Yang, N. Zhang, U. Iqbal, An in-depth investigation
1019 of data collection in llm app ecosystems, WWW '25, Association for
1020 Computing Machinery, 2025.
- 1021 [11] X. Hou, Y. Zhao, H. Wang, On the (In)Security of LLM App
1022 Stores , in: 2025 IEEE Symposium on Security and Privacy (SP),
1023 IEEE Computer Society, Los Alamitos, CA, USA, 2025, pp. 317–335.
1024 doi:10.1109/SP61157.2025.00117.
1025 URL [https://doi.ieeecomputersociety.org/10.1109/SP61157.](https://doi.ieeecomputersociety.org/10.1109/SP61157.2025.00117)
1026 2025.00117
- 1027 [12] OpenAI, Introducing GPTs, <https://openai.com/blog/introducing-gpts> (2023).
1028
- 1029 [13] P. Newswire, FlowGPT, The Open Ecosystem AI Platform, Raises
1030 \$10M, [https://www.prnewswire.com/news-releases/flowgpt-t](https://www.prnewswire.com/news-releases/flowgpt-the-open-ecosystem-ai-platform-raises-10m-302070574.html)
1031 [he-open-ecosystem-ai-platform-raises-10m-302070574.html](https://www.prnewswire.com/news-releases/flowgpt-the-open-ecosystem-ai-platform-raises-10m-302070574.html)
1032 (2024).
- 1033 [14] SimilarWeb, Website Analysis: poe.com, Available online, [https://pr](https://pro.similarweb.com/)
1034 [o.similarweb.com/](https://pro.similarweb.com/), Accessed: 2025-08-12 (2025).
- 1035 [15] SimilarWeb, Website Analysis: huggingface.co, Available online, [https:](https://pro.similarweb.com/)
1036 [//pro.similarweb.com/](https://pro.similarweb.com/), Accessed: 2025-08-12 (2025).
- 1037 [16] GPTStore.AI, GPTStore.AI, <https://gptstore.ai/gpts> (2024).
- 1038 [17] G. Hunter, GPTs Hunter, <https://www.gptshunter.com/> (2024).
- 1039 [18] N. Chen, S. C. Hoi, S. Li, X. Xiao, Mobile app tagging, in: Proceedings
1040 of the Ninth ACM International Conference on Web Search and Data
1041 Mining, WSDM '16, Association for Computing Machinery, New York,
1042 NY, USA, 2016, p. 63–72. doi:10.1145/2835776.2835812.
1043 URL <https://doi.org/10.1145/2835776.2835812>
- 1044 [19] M. H. Maqbool, U. Farooq, A. Mosharrof, A. B. Siddique, H. Foroosh,
1045 Mobilerec: A large scale dataset for mobile apps recommendation,
1046 in: Proceedings of the 46th International ACM SIGIR Conference on
1047 Research and Development in Information Retrieval, SIGIR '23, As-
1048 sociation for Computing Machinery, New York, NY, USA, 2023, p.

1049 3007–3016. doi:10.1145/3539618.3591906.
1050 URL <https://doi.org/10.1145/3539618.3591906>

1051 [20] OpenAI, Has anyone else noticed changes in the gpts search ranking
1052 algorithm lately?, Available online, [https://community.openai.com](https://community.openai.com/t/has-anyone-else-noticed-changes-in-the-gpts-search-ranking-algorithm-lately/1258050)
1053 [/t/has-anyone-else-noticed-changes-in-the-gpts-search-rank](https://community.openai.com/t/has-anyone-else-noticed-changes-in-the-gpts-search-ranking-algorithm-lately/1258050)
1054 [ing-algorithm-lately/1258050](https://community.openai.com/t/has-anyone-else-noticed-changes-in-the-gpts-search-ranking-algorithm-lately/1258050), Accessed: 2025-08-12 (2024).

1055 [21] OpenAI, Gpt store search change step back, Available on-
1056 line, [https://community.openai.com/t/gptstore-search-change-](https://community.openai.com/t/gptstore-search-change-step-back/600782)
1057 [step-back/600782](https://community.openai.com/t/gptstore-search-change-step-back/600782), Accessed: 2025-08-12 (2024).

1058 [22] OpenAI, Ranking of top 500 gpts by # conversations, Available on-
1059 line, [https://community.openai.com/t/ranking-of-top-500-gpts](https://community.openai.com/t/ranking-of-top-500-gpts-by-conversations/578607)
1060 [-by-conversations/578607](https://community.openai.com/t/ranking-of-top-500-gpts-by-conversations/578607), Accessed: 2025-08-12 (2024).

1061 [23] Q. Zheng, X. Xia, X. Zou, Y. Dong, S. Wang, Y. Xue, L. Shen, Z. Wang,
1062 A. Wang, Y. Li, et al., Codegeex: A pre-trained model for code gener-
1063 ation with multilingual benchmarking on humaneval-x, in: Proceedings
1064 of the 29th ACM SIGKDD Conference on Knowledge Discovery and
1065 Data Mining, 2023, pp. 5673–5684.

1066 [24] Y. Wang, X. Ma, G. Zhang, Y. Ni, A. Chandra, S. Guo, W. Ren, A. Arul-
1067 raj, X. He, Z. Jiang, et al., Mmlu-pro: A more robust and challenging
1068 multi-task language understanding benchmark, Advances in Neural In-
1069 formation Processing Systems 37 (2024) 95266–95290.

1070 [25] J. Yao, X. Yi, X. Xie, Clave: An adaptive framework for evaluating
1071 values of llm generated responses, in: A. Globerson, L. Mackey, D. Bel-
1072 grave, A. Fan, U. Paquet, J. Tomczak, C. Zhang (Eds.), Advances in
1073 Neural Information Processing Systems, Vol. 37, Curran Associates,
1074 Inc., 2024, pp. 58868–58900.
1075 URL [https://proceedings.neurips.cc/paper_files/paper/2024/](https://proceedings.neurips.cc/paper_files/paper/2024/file/6c1d2496c04d1ef648d58684b699643f-Paper-Datasets_and_Benchmarks_Track.pdf)
1076 [file/6c1d2496c04d1ef648d58684b699643f-Paper-Datasets_and_B](https://proceedings.neurips.cc/paper_files/paper/2024/file/6c1d2496c04d1ef648d58684b699643f-Paper-Datasets_and_Benchmarks_Track.pdf)
1077 [enchmarks_Track.pdf](https://proceedings.neurips.cc/paper_files/paper/2024/file/6c1d2496c04d1ef648d58684b699643f-Paper-Datasets_and_Benchmarks_Track.pdf)

1078 [26] N. Guha, J. Nyarko, D. Ho, C. Ré, A. Chilton, A. Chohlas-Wood, A. Pe-
1079 ters, B. Waldon, D. Rockmore, D. Zambrano, et al., Legalbench: A
1080 collaboratively built benchmark for measuring legal reasoning in large

- 1081 language models, *Advances in neural information processing systems* 36
1082 (2023) 44123–44279.
- 1083 [27] H. Rao, Y. Zhao, X. Hou, S. Wang, H. Wang, Software engineering for
1084 large language models: Research status, challenges and the road ahead,
1085 arXiv preprint arXiv:2506.23762 (2025).
- 1086 [28] D. Su, Y. Zhao, X. Hou, S. Wang, H. Wang, Gpt store mining and
1087 analysis, in: *Proceedings of the 16th International Conference on Inter-*
1088 *netware*, 2025, pp. 344–354.
- 1089 [29] Z. Zhang, L. Zhang, X. Yuan, A. Zhang, M. Xu, F. Qian, A first look
1090 at gpt apps: Landscape and vulnerability (2024). arXiv:2402.15105.
1091 URL <https://arxiv.org/abs/2402.15105>
- 1092 [30] C.-H. Chiang, H.-y. Lee, Can large language models be an alternative
1093 to human evaluations?, arXiv preprint arXiv:2305.01937 (2023).
- 1094 [31] J. Yang, J. Yang, K. Jin, Y. Miao, L. Zhang, L. Yang, Z. Cui, Y. Zhang,
1095 B. Hui, J. Lin, Evaluating and aligning codellms on human preference,
1096 arXiv preprint arXiv:2412.05210 (2024).
- 1097 [32] C. Grévisse, Llm-based automatic short answer grading in undergradu-
1098 ate medical education, *BMC Medical Education* 24 (1) (2024) 1060.
- 1099 [33] J. Gu, X. Jiang, Z. Shi, H. Tan, X. Zhai, C. Xu, W. Li, Y. Shen,
1100 S. Ma, H. Liu, et al., A survey on llm-as-a-judge, arXiv preprint
1101 arXiv:2411.15594 (2024).
- 1102 [34] D. Su, Y. Zhao, X. Hou, S. Wang, H. Wang, Gpt store mining and
1103 analysis, arXiv preprint arXiv:2405.10210 (2024).
- 1104 [35] P. Schober, C. Boer, L. A. Schwarte, Correlation coefficients: appro-
1105 priate use and interpretation, *Anesthesia & analgesia* 126 (5) (2018)
1106 1763–1768.
- 1107 [36] Y. Cao, L. Zhou, S. Lee, L. Cabello, M. Chen, D. Hershcovich, Assess-
1108 ing cross-cultural alignment between chatgpt and human societies: An
1109 empirical study, arXiv preprint arXiv:2303.17466 (2023).

- 1110 [37] X. Bai, A. Wang, I. Sucholutsky, T. L. Griffiths, Measuring im-
 1111 plicit bias in explicitly unbiased large language models, arXiv preprint
 1112 arXiv:2402.04105 (2024).
- 1113 [38] X. Hou, Y. Zhao, S. Wang, H. Wang, Gptzoo: A large-scale dataset
 1114 of gpts for the research community, in: Proceedings of the 39th
 1115 IEEE/ACM International Conference on Automated Software Engineer-
 1116 ing, ASE '24, Association for Computing Machinery, New York, NY,
 1117 USA, 2024, p. 2329–2333. doi:10.1145/3691620.3695309.
 1118 URL <https://doi.org/10.1145/3691620.3695309>
- 1119 [39] Y. Xie, X. Hou, Y. Zhao, K. Chen, H. Wang, Llm app squatting and
 1120 cloning, in: Proceedings of the 33rd ACM International Conference on
 1121 the Foundations of Software Engineering, 2025, pp. 64–74.
- 1122 [40] J. Krolik, H. Mahal, F. Ahmad, G. Trivedi, B. Saket, Towards leveraging
 1123 large language models for automated medical q&a evaluation, arXiv
 1124 preprint arXiv:2409.01941 (2024).
- 1125 [41] Z. A. Nazi, W. Peng, Large language models in healthcare and medical
 1126 domain: A review, in: Informatics, Vol. 11, MDPI, 2024, p. 57.
- 1127 [42] Q. Wei, Z. Yao, Y. Cui, B. Wei, Z. Jin, X. Xu, Evaluation of chatgpt-
 1128 generated medical responses: a systematic review and meta-analysis,
 1129 Journal of biomedical informatics 151 (2024) 104620.
- 1130 [43] Y. Guo, Z. Xu, Y. Yang, Is chatgpt a financial expert? evaluating
 1131 language models on financial natural language processing, arXiv preprint
 1132 arXiv:2310.12664 (2023).
- 1133 [44] Q. Xie, W. Han, Z. Chen, R. Xiang, X. Zhang, Y. He, M. Xiao, D. Li,
 1134 Y. Dai, D. Feng, et al., Finben: A holistic financial benchmark for large
 1135 language models, Advances in Neural Information Processing Systems
 1136 37 (2024) 95716–95743.
- 1137 [45] C. Xiao, S. X. Xu, K. Zhang, Y. Wang, L. Xia, Evaluating reading
 1138 comprehension exercises generated by llms: A showcase of chatgpt in
 1139 education applications, in: Proceedings of the 18th workshop on inno-
 1140 vative use of NLP for building educational applications (BEA 2023),
 1141 2023, pp. 610–625.

- 1142 [46] A. Kavadella, M. A. D. Da Silva, E. G. Kaklamanos, V. Stamatopoulos,
1143 K. Giannakopoulos, et al., Evaluation of chatgpt’s real-life implemen-
1144 tation in undergraduate dental education: mixed methods study, *JMIR*
1145 *Medical Education* 10 (1) (2024) e51344.
- 1146 [47] J. Wang, H. Zhao, Z. Yang, P. Shu, J. Chen, H. Sun, R. Liang, S. Li,
1147 P. Shi, L. Ma, et al., Legal evaluations and challenges of large language
1148 models, *arXiv preprint arXiv:2411.10137* (2024).
- 1149 [48] H. Li, J. Chen, J. Yang, Q. Ai, W. Jia, Y. Liu, K. Lin, Y. Wu, G. Yuan,
1150 Y. Hu, et al., Legalagentbench: Evaluating llm agents in legal domain,
1151 *arXiv preprint arXiv:2412.17259* (2024).
- 1152 [49] H. Li, Y. Chen, Q. Ai, Y. Wu, R. Zhang, Y. Liu, Lexeval: A compre-
1153 hensive chinese legal benchmark for evaluating large language models,
1154 *Advances in Neural Information Processing Systems* 37 (2024) 25061–
1155 25094.
- 1156 [50] J. Savelka, Unlocking practical applications in legal domain: Evaluation
1157 of gpt for zero-shot semantic annotation of legal texts, in: *Proceedings*
1158 *of the Nineteenth International Conference on Artificial Intelligence and*
1159 *Law*, 2023, pp. 447–451.
- 1160 [51] K. Papineni, S. Roukos, T. Ward, W.-J. Zhu, Bleu: a method for au-
1161 tomatic evaluation of machine translation, in: *Proceedings of the 40th*
1162 *annual meeting of the Association for Computational Linguistics*, 2002,
1163 pp. 311–318.
- 1164 [52] C.-Y. Lin, Rouge: A package for automatic evaluation of summaries, in:
1165 *Text summarization branches out*, 2004, pp. 74–81.
- 1166 [53] S. Banerjee, A. Lavie, Meteor: An automatic metric for mt evaluation
1167 with improved correlation with human judgments, in: *Proceedings of the*
1168 *acl workshop on intrinsic and extrinsic evaluation measures for machine*
1169 *translation and/or summarization*, 2005, pp. 65–72.
- 1170 [54] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, Y. Artzi, Bertscore:
1171 Evaluating text generation with bert, *arXiv preprint arXiv:1904.09675*
1172 (2019).

- 1173 [55] T. Sellam, D. Das, A. P. Parikh, Bleurt: Learning robust metrics for
1174 text generation, arXiv preprint arXiv:2004.04696 (2020).
- 1175 [56] C. Dong, Y. Li, H. Gong, M. Chen, J. Li, Y. Shen, M. Yang, A survey
1176 of natural language generation, ACM Computing Surveys 55 (8) (2022)
1177 1–38.
- 1178 [57] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin,
1179 C. Zhang, S. Agarwal, K. Slama, A. Ray, et al., Training language mod-
1180 els to follow instructions with human feedback, Advances in neural in-
1181 formation processing systems 35 (2022) 27730–27744.
- 1182 [58] W.-L. Chiang, L. Zheng, Y. Sheng, A. N. Angelopoulos, T. Li, D. Li,
1183 B. Zhu, H. Zhang, M. Jordan, J. E. Gonzalez, et al., Chatbot arena: An
1184 open platform for evaluating llms by human preference, in: Forty-first
1185 International Conference on Machine Learning, 2024.
- 1186 [59] S. Shankar, J. Zamfirescu-Pereira, B. Hartmann, A. Parameswaran,
1187 I. Arawjo, Who validates the validators? aligning llm-assisted evalua-
1188 tion of llm outputs with human preferences, in: Proceedings of the 37th
1189 Annual ACM Symposium on User Interface Software and Technology,
1190 2024, pp. 1–14.